

Untitled5

May 19, 2026

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[2]: import pandas as pd
import numpy as np
import statsmodels.api as sm
import statsmodels.formula.api as smf

# 1. Load the official dataset online from UCLA server
url_nb = "https://stats.idre.ucla.edu/stat/stata/dae/nb_data.dta"
df_nb = pd.read_stata(url_nb)

# Rename columns to match standard formula naming if needed, and set category
df_nb['prog'] = df_nb['prog'].astype('category')

# 2. Fit Negative Binomial Model using statsmodels GLM
# The default uses log link. We pass the count data directly.
nb_model = smf.glm(formula="daysabs ~ math + C(prog)", data=df_nb,
                    family=sm.families.NegativeBinomial()).fit()

# 3. Display summary
print(nb_model.summary())

# 4. Compute Incidence Rate Ratios (IRR) and 95% Confidence Intervals
params_nb = nb_model.params
conf_nb = nb_model.conf_int()
irr_nb = np.exp(params_nb)
conf_irr_nb = np.exp(conf_nb)

# Structure results into a clean summary DataFrame
nb_results = pd.DataFrame({
    'Coefficient (Beta)': params_nb,
    'Incidence Rate Ratio (IRR)': irr_nb,
    'Lower CI (IRR)': conf_irr_nb[0],
    'Upper CI (IRR)': conf_irr_nb[1],
    'p-value': nb_model.pvalues
})
print("\n===== PROBLEM 7: ONLINE CALCULATED IRR AND 95% CI =====")
print(nb_results.round(4))
```

Generalized Linear Model Regression Results

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Dep. Variable:          daysabs    No. Observations:          314
Model:                  GLM        Df Residuals:              310
Model Family:          NegativeBinomial    Df Model:                3
Link Function:          Log         Scale:                  1.0000
Method:                 IRLS        Log-Likelihood:        -865.68
Date:                   Tue, 19 May 2026    Deviance:              350.98
Time:                   00:37:57           Pearson chi2:          331.
No. Iterations:         6              Pseudo R-squ. (CS):      0.1926
Covariance Type:        nonrobust
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==
              coef      std err          z      P>|z|      [0.025
0.975]
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Intercept          2.6150      0.200     13.054      0.000      2.222
3.008
C(prog)[T.2.0]     -0.4408      0.185     -2.379      0.017     -0.804
-0.078
C(prog)[T.3.0]     -1.2786      0.203     -6.284      0.000     -1.677
-0.880
math              -0.0060      0.003     -2.358      0.018     -0.011
-0.001
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===== PROBLEM 7: ONLINE CALCULATED IRR AND 95% CI =====

	Coefficient (Beta)	Incidence Rate Ratio (IRR) \
Intercept	2.6150	13.6674
C(prog)[T.2.0]	-0.4408	0.6436
C(prog)[T.3.0]	-1.2786	0.2784
math	-0.0060	0.9940

	Lower CI (IRR)	Upper CI (IRR)	p-value
Intercept	9.2293	20.2396	0.0000
C(prog)[T.2.0]	0.4476	0.9253	0.0174
C(prog)[T.3.0]	0.1868	0.4149	0.0000
math	0.9891	0.9990	0.0184

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C:\Users\user\anaconda3\Lib\site-
packages\statsmodels\genmod\families\family.py:1367: ValueWarning: Negative
binomial dispersion parameter alpha not set. Using default value alpha=1.0.
warnings.warn("Negative binomial dispersion parameter alpha not "

```

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3. Statistical Interpretation and Findings

Based on the Negative Binomial regression output ($\alpha = 0.05$), the findings are summarized below:

- **Math Score** ($p = 0.018$): Significant negative effect. Every 1-unit increase in math score reduces the expected student absence rate by **0.60%** (IRR = 0.9940).
- **Academic Track** ($p = 0.017$): Compared to General track, Academic students have a **35.64% lower expected rate** of absences (IRR = 0.6436).
- **Vocational Track** ($p < 0.001$): Highly significant. Vocational placement reduces the expected absence rate by a massive **72.16%** (IRR = 0.2784).